Aethera — University Edition

Unified Modelling Language (UML) Software Engineering Final Report

Course: Software Engineering & UML | Author: Omar Saeed Mohamed Kamel | ID: AI8 - G3 - 0543 | Model: Llama 3.3 70B via Groq API | Status: Complete Formal Deliverable

1. Introduction & Executive Summary

Aethera (University Edition) is an advanced, AI-powered academic assistant designed to optimize study workflows, research synthesis, exam preparation, and knowledge retention for university students. Operating as a hybrid Single Page Application (SPA) with a Node.js + Express backend and SQLite/Supabase PostgreSQL databases, Aethera leverages the ultra-fast **Groq Cloud AI Infrastructure (Llama 3.3 70B Versatile model)** to deliver instantaneous academic tools.

This report documents the end-to-end software engineering analysis, requirements specifications, and comprehensive UML object-oriented design suite (all 8 required diagrams) for the platform.

1.1 Project Objectives

🎓 Academic Productivity

Provide automated summarization, question generation, concept breakdown,

⚡ Zero-Latency AI Access

Integrate directly with Groq API Llama 3.3 70B to achieve response rendering in under 1.5 seconds.

and essay assistance tailored to higher education standards.

🔄 Multi-System Integration

Support direct synchronization with Notion API, Anki flashcard deck formats, local PDF/DOCX document parsers, and SQLite/PostgreSQL storage.

📐 Full UML Engineering

Model all structural, behavioral, data, and interaction aspects using standard UML diagrams.

2. Software Requirements Specification (SRS)

2.1 Functional Requirements (FR)

ID	Module / Feature	Requirement Description	Priority
FR-01	Academic Summarizer	System shall summarize academic texts, lecture notes, and PDFs using 4 distinct styles (Brief, Detailed, Critical Review, Executive).	High
FR-02	Exam Generator	System shall generate practice exams across 5 question types (MCQ, Essay, Short Answer, True/False, Case Study) with 4 difficulty levels.	High
FR-03	Concept Deep Dive	System shall break down complex academic terms into procedural definitions, historical origins, key scholars, and common misconceptions.	Medium
FR-04	Smart Study Planner	System shall construct tailored study schedules based on exam dates, daily available hours, and learning	High

ID	Module / Feature	Requirement Description	Priority
		techniques (Pomodoro, Spaced Repetition).	
FR-05	Flashcard Generator	System shall convert input text into interactive flip cards with shuffle, progress tracking, and CSV/Anki export.	High
FR-06	Mind Map Visualizer	System shall render hierarchical tree mind maps in vector SVG format directly from topic prompts.	High
FR-07	Quick Quiz Engine	System shall generate self-correcting interactive quizzes with real-time scoring and answer explanations.	High
FR-08	Citation Builder	System shall auto-generate academic citations in 6 standard formats (APA 7th, MLA 9th, Chicago, Harvard, IEEE, Vancouver).	Medium
FR-09	GPA Calculator	System shall compute Semester and Cumulative GPA supporting Egyptian & US grading scales with dynamic semester additions.	Medium
FR-10	Notion Live Sync	System shall export generated notes, summaries, and flashcards directly into user's Notion workspace database via API.	High
FR-11	Document Parser	System shall extract raw text locally from PDF, TXT, and DOCX files without uploading sensitive student files to external storage.	High
FR-12	History & Storage	System shall persist study sessions, AI responses, notes, and preferences in LocalStorage and SQLite database.	High

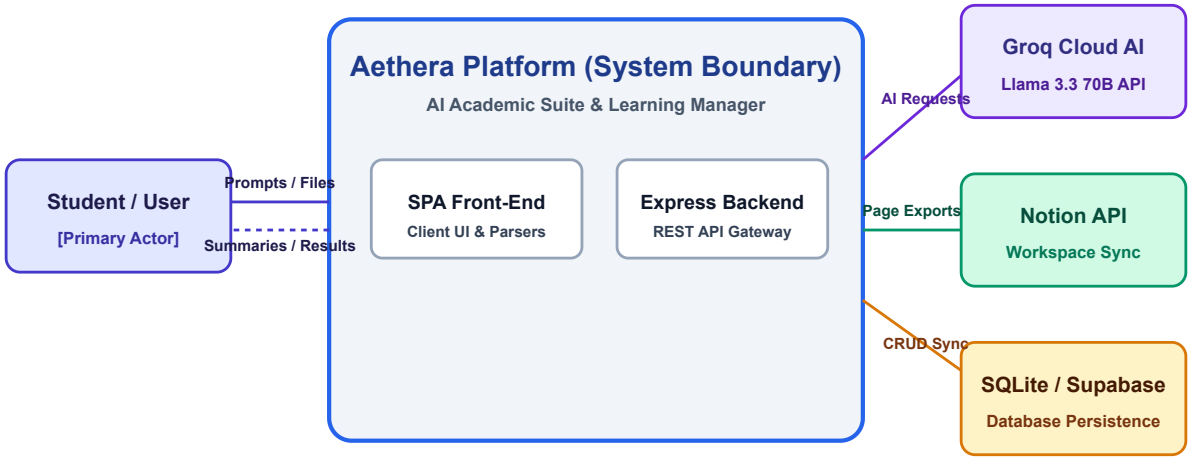
2.2 Non-Functional Requirements (NFR)

ID	Category	Requirement Description
NFR-01	Performance	The UI shall render tools instantly (< 100ms) and complete AI stream processing within 2.0 seconds for average prompts.
NFR-02	Security	Groq API keys and Notion tokens shall be stored locally on client browser storage and never exposed in plain text logs.
NFR-03	Usability	The UI shall implement Cyberpunk Dark Glassmorphism aesthetic with full support for RTL (Arabic) and LTR (English) layouts.
NFR-04	Reliability	The application shall operate seamlessly offline for local tools (GPA Calculator, Local Storage History, Pomodoro Timer).
NFR-05	Portability	The SPA prototype shall run on any modern Evergreen browser (Chrome, Edge, Firefox, Safari) without external software installation.

3. Complete UML Diagrams & Architectural Specifications

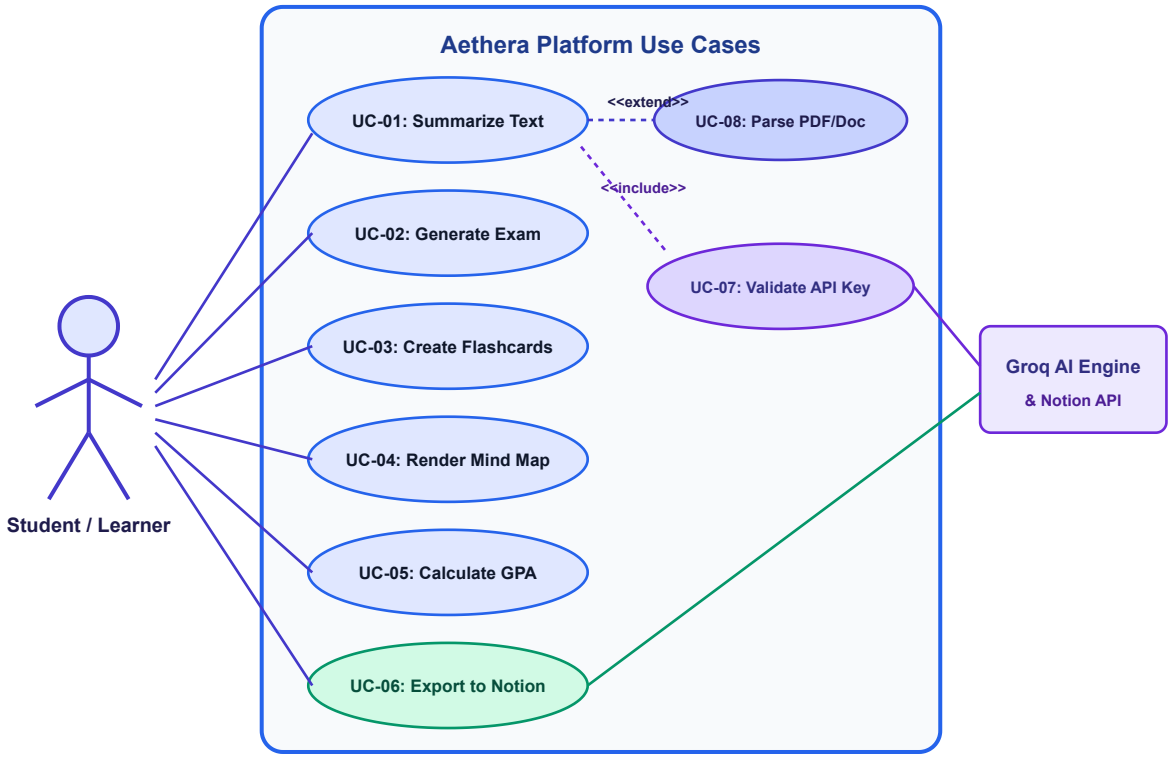
[1] Context Diagram

Defines system boundaries and external entity interactions with the Aethera Platform.


Aethera Platform (System Boundary)
AI Academic Suite & Learning Manager
Student / User
[Primary Actor]
Prompts / Files
Summaries / Results
SPA Front-End
Client UI & Parsers
Express Backend
REST API Gateway
AI Requests
Groq Cloud AI
Llama 3.3 70B API
Page Exports
Notion API
Workspace Sync
CRUD Sync
SQLite / Supabase
Database Persistence

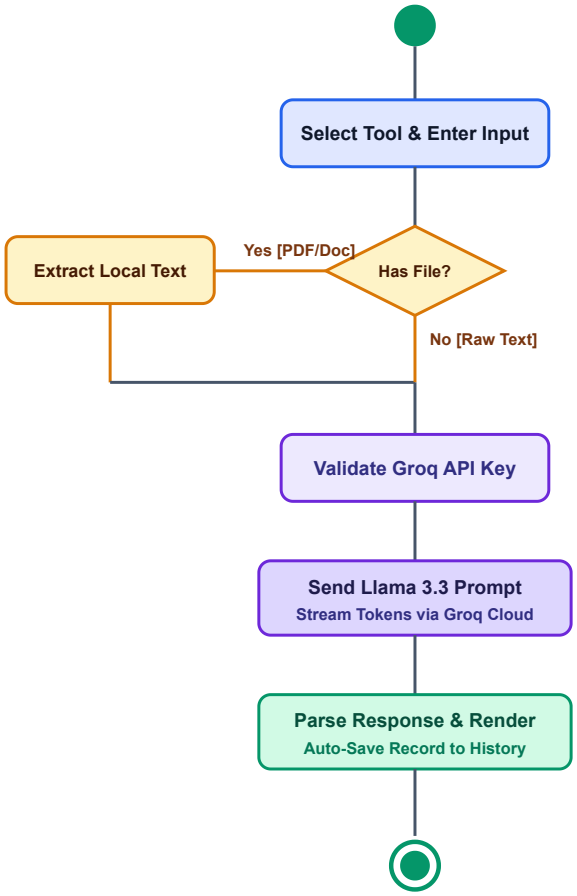
[2] Use Case Diagram

Illustrates system capabilities from user perspective with actor relationships and inclusion/extension dependencies.



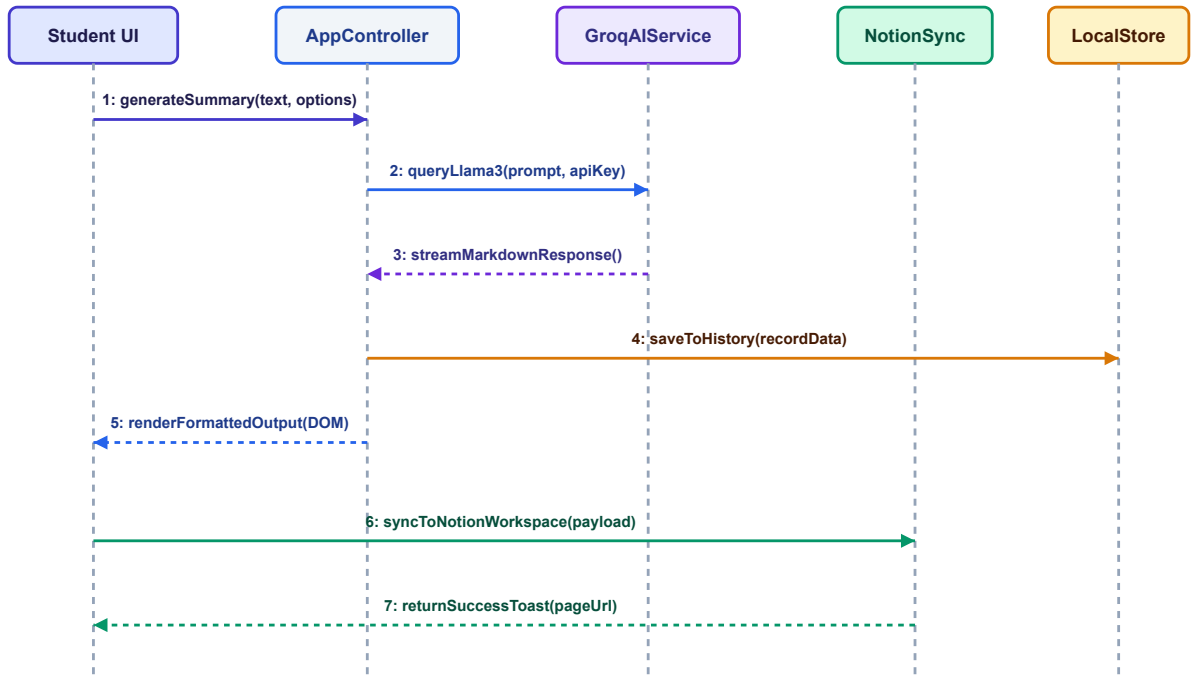
[3] Activity Diagram

Demonstrates procedural workflow execution from user submission to parsing, API stream processing, and history persistence.


Select Tool & Enter Input
Has File?
Yes [PDF/Doc]
Extract Local Text
No [Raw Text]
Validate Groq API Key
Send Llama 3.3 Prompt
Stream Tokens via Groq Cloud
Parse Response & Render
Auto-Save Record to History

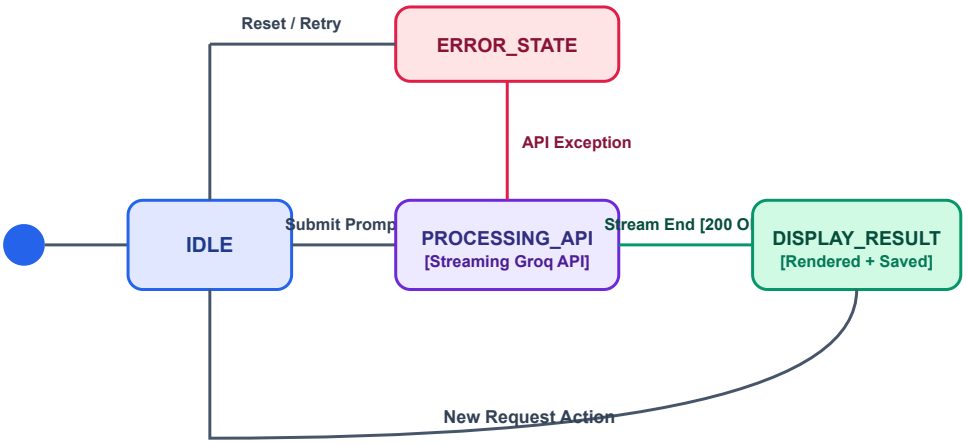
[4] Sequence Diagram

Sequence of object interactions and message exchanges across system components during prompt processing.



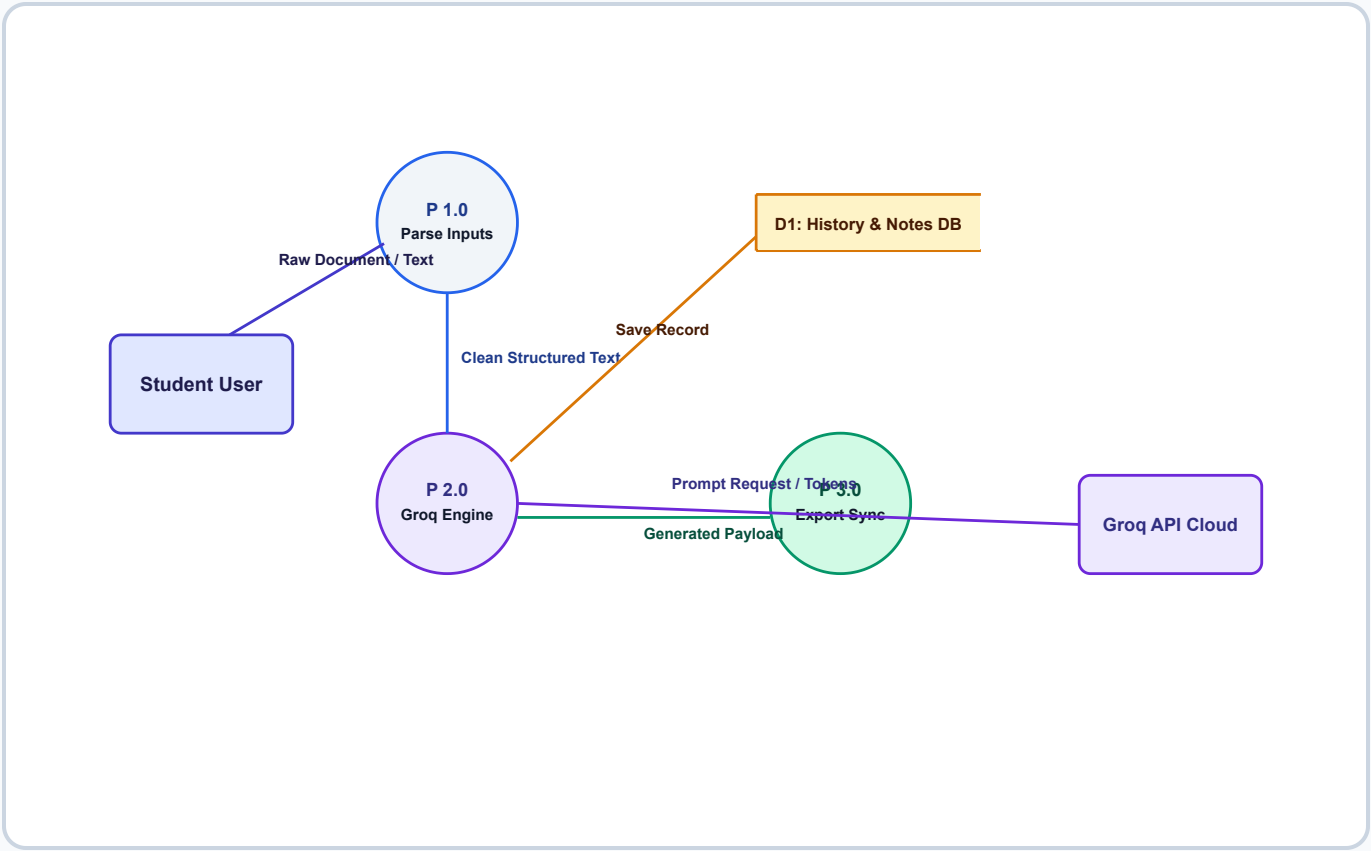
[5] State Machine Diagram

Models behavioral state transitions of the Aethera platform session life cycle.


Reset / Retry
ERROR_STATE
API Exception
IDLE
Submit Promp
PROCESSING_API
[Streaming Groq API]
Stream End [200 O
DISPLAY_RESULT
[Rendered + Saved]
New Request Action

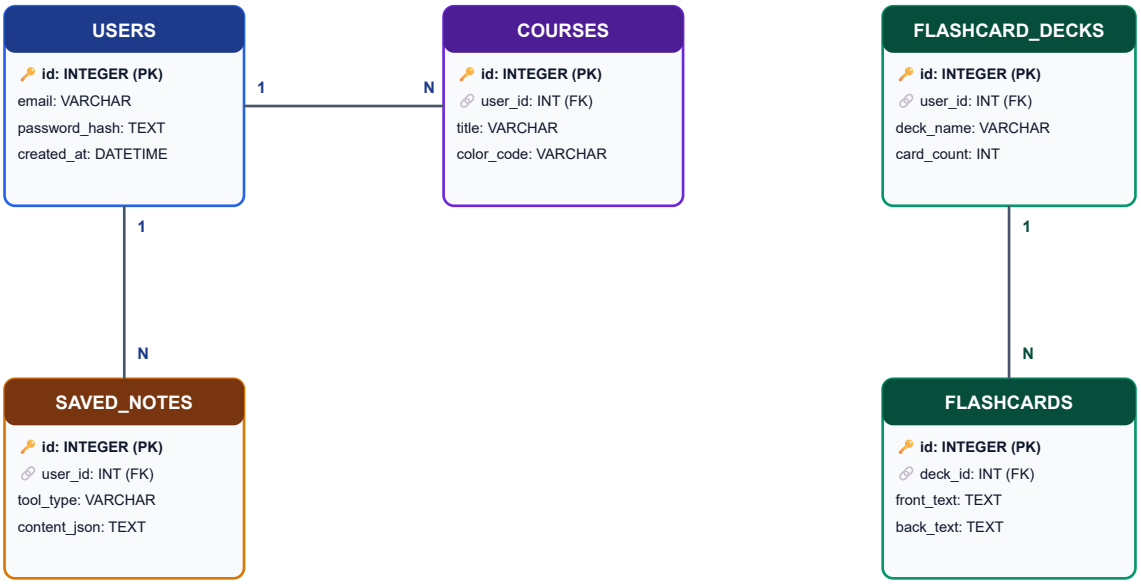
[6] Data Flow Diagram (DFD Level 1)

Models functional processes, data transformations, and storage repositories within the system.



[7] Entity Relationship Diagram (ERD)

Data structure model representing the 17 core entities, primary keys, and relationships.


USERS
id: INTEGER (PK)
email: VARCHAR
password_hash: TEXT
created_at: DATETIME
1
N
COURSES
id: INTEGER (PK)
user_id: INT (FK)
title: VARCHAR
color_code: VARCHAR
FLASHCARD_DECKS
id: INTEGER (PK)
user_id: INT (FK)
deck_name: VARCHAR
card_count: INT
1
N
1
N
SAVED_NOTES
id: INTEGER (PK)
user_id: INT (FK)
tool_type: VARCHAR
content_json: TEXT
FLASHCARDS
id: INTEGER (PK)
deck_id: INT (FK)
front_text: TEXT
back_text: TEXT

[8] Class Diagram

Object-oriented structural design detailing class attributes, methods, visibility, and associations.

